



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

12-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B	C,D	A,B,D	A,C	A,B	A,C	A,C	A,B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C,D	A,D	B,D	A,D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.64	16.00	40.32	6.67	0.20	90.01	40.00	10.00 to 10.05

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,D	A,B,C	C,D	A,D	A,C	A	B,C	B,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A,C	B	A,B,C,D	A,B,C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	7.00	5.00	7.00	5.00	0.02	2.77	2.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B,D	B,C,D	B,C	A	A,B,C,D	A,C	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A,B,C	A,C	B,C,D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.50	6.00	8.00	4.58 to 4.59	0.03	0.25	3.14	0.50

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B)

If $x < a_0$

$$\frac{2e^2 x}{4\pi\epsilon_0 a_0^3} = \frac{mv^2}{x}$$

$$\frac{nh}{2\pi} = mvx$$

$$v = \frac{ex}{\sqrt{2\pi\epsilon_0 a_0^3 m}}$$

$$\frac{nh}{2\pi} = \frac{mex^2}{\sqrt{2\pi\epsilon_0 a_0^3 m}}$$

$$x = \sqrt{\frac{nh}{e} \sqrt{\frac{\epsilon_0 a_0^3}{2\pi m}}} < a_0$$

$$a_0 = \frac{\epsilon_0 h^2}{\pi m e^2}$$

$$\frac{n^2 h^2}{e^2} \times \frac{\epsilon_0 a_0^3}{2\pi m} < a_0^4 = \frac{\epsilon_0 h^2}{\pi m e^2}$$

$$n^2 < 2$$

$$\Rightarrow n = 1$$

$$v = \frac{e}{\sqrt{2\pi m \epsilon_0 a_0^3}} \times \sqrt{\frac{h}{e} \sqrt{\frac{\epsilon_0 a_0^3}{2\pi m}}} = \sqrt{\frac{he}{2\pi m v^{3/2} \epsilon_0 a_0^3}}$$

8. Ans (A,B,C,D)

$$\Delta\phi = \frac{2\pi}{4 \times 10^{-7}} \left(0.5 \times 10^{-6} - \frac{10^{-3} \times y}{2} \right)$$

PART-1 : PHYSICS

SECTION-I (ii)

12. Ans (A,D)

$$y = \frac{\omega \ell^3}{4bd^3\delta} = \frac{10 \times 1^3}{4 \times 0.1 \times 10^{-6} \times 10^{-3}}$$

$$= 2.5 \times 10^{10} \text{ N/m}^2$$

$$\frac{\Delta y}{y} = \frac{\Delta \omega}{\omega} + \frac{3\Delta \ell}{\ell} + \frac{\Delta b}{b} + \frac{3\Delta d}{d} + \frac{\Delta \delta}{\delta}$$

$$\frac{\Delta y}{2.5 \times 10^{10}} = \frac{1}{100} + 3 \times \frac{1}{100} + \frac{1}{100} + 3 \times \frac{1}{100} + \frac{1}{100}$$

$$= \frac{1+3+1+3+1}{100} = \frac{9}{100}$$

$$\Delta y = 2.25 \times 10^9 \text{ N/m}^2$$

PART-1 : PHYSICS

SECTION-II

1. Ans (0.64)

$$\ell = \frac{2\lambda f}{\omega}$$

$$\omega = \frac{2\lambda f}{\ell} = \frac{2 \times 420 \times 10^{-9} \times 0.8}{1.05 \times 10^{-3}} = 0.64 \text{ mm}$$

2. Ans (16.00)

$$2\mu t = \lambda$$

$$t = \frac{480 \times 10^{-9}}{3} = 160 \text{ nm}$$

4. Ans (6.67)

$$n = N_0(1 - e^{-2\lambda}) = N_0(1 - x)$$

$$\frac{7n}{4} = N_0(1 - e^{-4\lambda}) = N_0(1 - x^2)$$

$$\frac{7}{4} = 1 + x$$

$$x = \frac{3}{4} = e^{-2\lambda}$$

$$\frac{\ln 4 - \ln 3}{2} = \frac{1}{T}$$

$$T = \frac{2}{\ln 4 - \ln 3} = \frac{2}{1.4 - 1.1}$$

$$= \frac{20}{3} = 6.67$$

5. Ans (0.20)

$$P_1 = A\sigma(T_1^4 - T_2^4) = 6 \times 10^{-8} \times A(4^4 - 3^4) \times 10^8$$

$$= 1050 \text{ A watts}$$

$$P_2 = \frac{kA}{L}(T_1 - T_2) = \frac{2.1 \times A}{1} \times 100 = 210 \text{ A watts}$$

8. Ans (10.00 to 10.05)

$$k = \frac{hc}{\lambda} = \frac{1240}{310} = 4 \text{ eV}$$

$$E = 3.5 \text{ eV}$$

$$E = 3.5 - \frac{hc}{\lambda_0} = 3.5 - \frac{1240}{620} = 1.5 \text{ eV}$$

$$\lambda = \sqrt{\frac{150}{1.5}} = 10 \text{ Å}$$

PART-2 : CHEMISTRY

SECTION-I (i)

5. Ans (A,C)

Let the fractions of Cu present in (+1) oxidation state be 'x'.

$${}^a_{\text{Cu}}{}^{(-2)}_{1.8}\text{O}$$

$$\Rightarrow 1.8a - 2 = 0$$

$$\Rightarrow a = \left(\frac{2}{1.8}\right) = (x)(1) + (1-x)(2)$$

$$\Rightarrow x = \left(2 - \frac{2}{1.8}\right) = 2 \times \frac{0.8}{1.8} = \frac{1}{1.125}$$

6. Ans (A)

$$X_A = \frac{5}{25}, \quad X_B = \frac{20}{25}$$

$$= \frac{1}{5} \quad = \frac{4}{5}$$

Vapour pressure of solution should be

$$= \frac{1}{5} \times 25 + \frac{4}{5} \times 75$$

$$= 65 \text{ cm Hg}$$

Maintained pressure = 50 cm Hg

$\Rightarrow$ If all of A & B vaporise, then total pressure would be :

$$y_A = \frac{1}{5} = 0.2; y_A = 0.8$$

$$\Rightarrow \frac{1}{P_T} = \frac{0.2}{25} = \frac{0.8}{75}$$

$$\Rightarrow P_T = \frac{75}{1.4} = 53.57 \text{ cm Hg}$$

$\Rightarrow$ If maintained pressure is less than 53.57 cm Hg, entire liquid will vaporise.

PART-2 : CHEMISTRY

SECTION-I (ii)

11. **Ans (A,B,C,D)**

$$\begin{aligned}[X]_t &= [X]_0 e^{-k_1 t} \\ &= 0.5 \times e^{-0.05 \times 20} = 0.5 e^{-1} \\ &= 0.5 \times 0.36 = 0.18 \text{ M} \\ [Y]_t &= \frac{[X]_0 k_1}{(k_2 - k_1)} (e^{-k_1 t} - e^{-k_2 t}) \\ &= \frac{0.5 \times 0.05}{0.45} (e^{-1} - e^{-10}) \\ &\cong 0.02 \text{ M}\end{aligned}$$

At any instance;

$$\begin{aligned}[A]_0 &= [A]_t + [B]_t + [C]_t \\ \Rightarrow [C]_t &= [A]_0 - [A]_t - [B]_t \\ &= 0.5 - 0.18 - 0.02 = 0.3 \text{ M}\end{aligned}$$

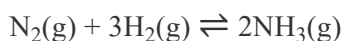
PART-2 : CHEMISTRY

SECTION-II

5. **Ans (5.00)**

FeCl₃-solution, Br₂-water, (KMnO₄ + H₂SO₄),
(starch-IO₃⁻), (K₂Cr₂O₇ + H₂SO₄)

6. **Ans (0.02)**



$$t = 0 \quad x \text{ mol} \quad 3x \text{ mol}$$

$$t = \infty \quad x-y \quad 3(x-y) \quad 2y$$

$$\text{Given } \frac{2y}{4x - 4y + 2y} = 0.2$$

$$\Rightarrow 10y = 4x - 2y$$

$$\Rightarrow x = 3y$$

$$\Rightarrow P_{\text{NH}_3} = \frac{2y}{4x - 2y} \times 100$$

$$= \frac{2y}{10y} \times 100 = 20 \text{ atm}$$

$$P_{\text{N}_2} = \frac{x - y}{10y} \times 100 = 20 \text{ atm}$$

$$\Rightarrow P_{\text{H}_2} = 60 \text{ atm}$$

$$\Rightarrow K_P = \frac{P_{\text{NH}_3}^2}{P_{\text{N}_2} P_{\text{H}_2}^3} = \frac{20 \times 20}{20 \times 60 \times 60 \times 60}$$

7. **Ans (2.77)**

For a reversible isothermal process;

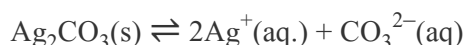
$$\begin{aligned}q = -w &= nRT \ln \left(\frac{V_2}{V_1} \right) \\ &= 2 \text{ mol} \times \frac{2 \text{ cal}}{\text{mol} \cdot \text{K}} \times 500 \text{ K} \ln \left(\frac{20}{5} \right) \\ &= 4 \times 0.693 \text{ kcal} = 2.772 \text{ kcal}\end{aligned}$$

8. **Ans (2.00)**

The above cell may be represented as



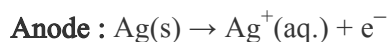
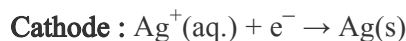
Where;



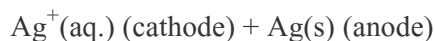
$$2s \quad 0.2$$

$$\Rightarrow 4s^2 \times 0.2 = K_{\text{sp}}$$

$$\Rightarrow s = \left(\frac{K_{\text{sp}}}{0.8} \right)^{1/2}$$



Cell reaction :



$$Q = \frac{[\text{Ag}^+]_{\text{Anode}}}{[\text{Ag}^+]_{\text{Cathode}}} = \frac{(K_{\text{sp}}/0.2)^{1/2}}{(0.1)}$$

$$\Rightarrow E_{\text{cell}} = 0 - \frac{0.06}{1} \log \left(\frac{(K_{\text{sp}}/0.2)^{1/2}}{0.1} \right) = 0.18$$

$$\Rightarrow \log \left(\frac{0.1}{(K_{\text{sp}}/0.2)^{1/2}} \right) = 3$$

$$\Rightarrow K_{\text{sp}} = 2 \times 10^{-9}$$

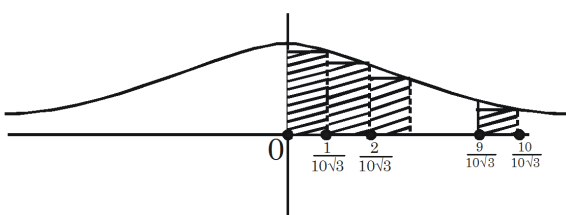
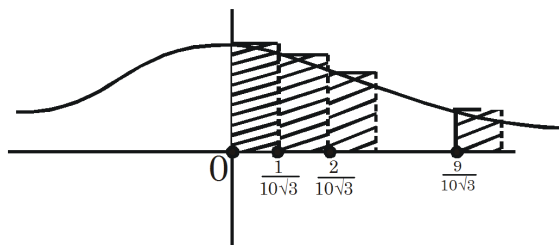
PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C)

$$f(t) = \frac{1}{1+t^2}$$

y is shaded area ←

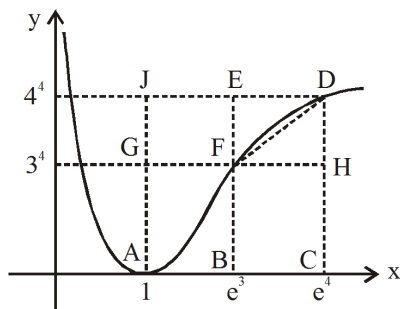


→ x is shaded area

$$y > \int_0^{1/\sqrt{3}} \frac{dt}{1+t^2} > x \Rightarrow x < \frac{\pi}{6} < y$$

$$\text{also } y - \frac{\pi}{6} < \frac{\pi}{6} - x \Rightarrow x + y < \frac{\pi}{3}$$

2. Ans (A,B,D)



A is area bounded by curve between $x=1, x=e^4$
 $x=e^3$ is point of inflection

Now, $A < \text{Ar.}(\square ABGF) + \text{Ar.}(\square BCDE)$

$$\Rightarrow A < (e^3 - 1)3^4 + (e^4 - e^3)4^4$$

$$\Rightarrow A < 256e^4 - 175e^3 - 81 \quad (\text{Ans. A})$$

Also, $A < \text{Ar.}(\square ACDJ)$

$$\Rightarrow A < (e^4 - 1)4^4$$

$$\Rightarrow A < 256(e^4 - 1) \quad (\text{Ans. B})$$

Again, $A > \text{Ar.}(\text{trapezium BFDC})$

$$\Rightarrow A > \frac{1}{2}(3^4 + 4^4)(e^4 - e^3)$$

$$\Rightarrow A > \frac{337}{2}(e^3(e - 1)) \quad (\text{Ans. D})$$

3. Ans (B,C,D)

Differentiate on both side

$$\frac{3z^3 - 8z + 5}{\sqrt{z^2 - 4z - 7}}$$

$$= (2z + a)\sqrt{z^2 - 4z - 7} + \frac{(z^2 + az + 36)}{\sqrt{z^2 - 4z - 7}}(z - 2) + \frac{b}{\sqrt{z^2 - 4z - 7}}$$

$$\Rightarrow 3z^3 - 8z + 5$$

$$= (2z + a)(z^2 - 4z - 7) + (z^2 + az + 36)(z - 2) + b$$

equating the coefficient of z^2 and constant

$$a - 8 + a - 2 = 0 \Rightarrow a = 5$$

$$5 = -7a - 72 + b \Rightarrow b = 112$$

4. Ans (B,C)

$$\left[\frac{x+1}{x} > 0 \right] \Rightarrow \left[x \in (-\infty, -1) \cup (0, \infty) \right] \quad \dots (1)$$

$$\left[\left| \ln \left(\frac{x+1}{x} \right) \right| \leq 1 \right] \Rightarrow \left[\frac{1}{e} \leq \frac{x+1}{x} \leq e \right] \quad \dots (2)$$

$$\text{From (2), } \frac{x+1-ex}{x} \leq 0 \quad \text{and}$$

$$\frac{ex + e - x}{ex} \geq 0$$

$$\Rightarrow \frac{x(e-1)-1}{x} \geq 0 \quad \text{and} \quad \frac{x(e-1)+e}{ex} \geq 0$$

$$x \in (-\infty, 0) \cup \left[\frac{1}{e-1}, \infty \right)$$

$$\text{and } x \in \left(-\infty, \frac{e}{1-e} \right] \cup (0, \infty)$$

from (1) and (2)

$$x \in \left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

and range of $f(x)$ is $[-1, 1]$

5. Ans (A)

$$f(2-x) = f(2+x)$$

$$x \rightarrow 2-x$$

$$f(x) = f(4-x) \quad \dots (i)$$

differentiate w.r.t. x

$$f'(x) = -f'(4-x) \quad \dots (ii)$$

Put $x = 1/2$ in (ii)

$$f'\left(\frac{1}{2}\right) = f'\left(\frac{7}{2}\right) = f'(1) = f'(2) = f'(3) = 0$$

also

$$\text{Put } x = 1, 2$$

Now consider $f'(x) = y$.

by Roll's theorem we can say in

$$\left[\frac{1}{2}, 1 \right], [1, 2], \left[2, \frac{7}{2} \right], \left[\frac{7}{2}, 3 \right]$$

$f''(x) = 0$ has minimum 4 roots.

6. Ans (A,B,C,D)

$$F(n, x) = \sum_{r=1}^n \tan^{-1} \left(\frac{r+1}{2} x \right) - \tan^{-1} \left(\frac{r}{2} \right) x$$

$$= \tan^{-1} \left(\frac{n+1}{2} x \right) - \tan^{-1} \left(\frac{x}{2} \right)$$

$$\Rightarrow G(n, x) = \frac{2(n+1)}{4 + (n+1)^2 x^2} - \frac{2}{4 + x^2}$$

$$G\left(n, \frac{1}{3}\right) = \frac{18(n+1)}{36 + (n+1)^2} - \frac{18}{37}$$

$$G\left(n, \frac{1}{3}\right) < 0 \Rightarrow n^2 > 35n \Rightarrow n \geq 36$$

7. Ans (A,C)

$$(a+b) + (b+c) + (a+c) = 8$$

$$\text{let } a+b = \alpha, b+c = \beta, a+c = \gamma$$

$$\Rightarrow \alpha + \beta + \gamma = 8$$

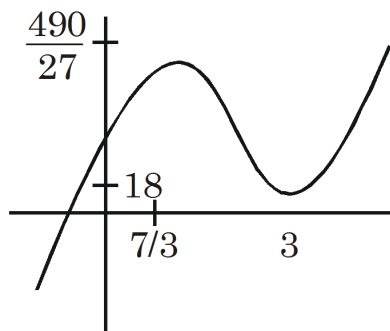
$$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = a^2 + b^2 + c^2 + 3(ab + bc + ac)$$

$$= 21$$

$$\Rightarrow x^3 - 8x^2 + 21x - \lambda = 0$$

$$\text{where } \lambda = (a+b)(b+c)(a+c)$$

these equation has 3 real roots

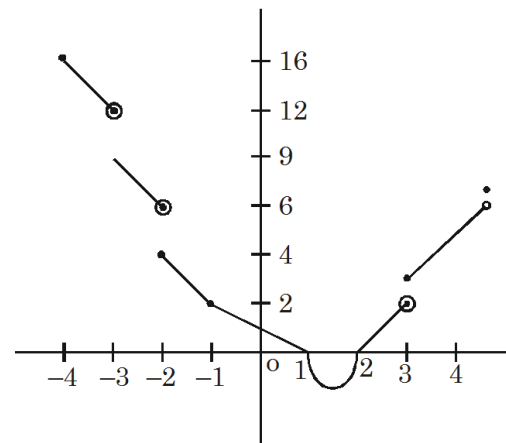


$$\text{so } f(3)f(7/3) \leq 0$$

$$18 \leq \lambda \leq \frac{490}{27}$$

8. Ans (A,B,C)

$$f(x) = \begin{cases} x[x] & -4 \leq x < -1 \\ 1-x & -1 \leq x < 1 \\ x^2 - 3x + 2 & 1 \leq x < 2 \\ (x-2)[x] & 2 \leq x \leq 4 \\ -4x & x \in [-4, -3] \\ -3x & x \in [-3, -2] \\ -2x & x \in [-2, -1] \\ 1-x & x \in [-1, 1) \\ x^2 - 3x + 2 & x \in [1, 2) \\ 2(x-2) & x \in [2, 3) \\ 3(x-2) & x \in [3, 4) \\ 8 & x = 4 \end{cases}$$



(1) $f(x)$ is DC at $x = -3, 2, 3, 4 \rightarrow 4$ points

(2) $f(x)$ is ND at $x = -3, -2, 3, 4, -1, 2 \rightarrow 6$ points

(3) minima at $x = \frac{3}{2}$, maxima at $x = -4, 4$

(4) no. of solution of $f(x) = 4$ is 2

(5) Sum of global maximum value and global minimum value of $\frac{63}{4}$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (A,B,C)

$$\int_0^x g(t) dt = 3x + \int_x^0 (\cos^2 t) g(t) dt$$

by leibnitz rule

$$g(x) = 3 - \cos^2 x \cdot g(x)$$

$$\text{or } g(x) = \frac{3}{1 + \cos^2 x}$$

$\therefore$ Maximum value of $g(x)$ is 3 at $\cos x = 0$

and minimum value of $g(x)$ is $3/2$ at $\cos x = 1$

$$\text{Now } 3 \int \frac{1}{1 + \cos^2 x} dx$$

$$\text{or } 3 \int_0^\infty \frac{\sec^2 x}{\tan^2 x + 2} dx$$

$$\text{or } \frac{3}{\sqrt{2}} \left[\tan^{-1} \left(\frac{\tan x}{\sqrt{2}} \right) \right]_0^\infty$$

$$\Rightarrow \frac{3}{\sqrt{2}} \cdot \frac{\pi}{2}$$

10. Ans (A,C)

$$f(x) = \lim_{\alpha \rightarrow 0} \frac{12}{\alpha^4} \int_0^x \frac{(e^{(x+t)} - e^x) \ln^2(1+t)}{(2t^3 + 3)} dt$$

by leibnitz rule differentiate w.r.t. ' α '

$$f(x) = \lim_{\alpha \rightarrow 0} \frac{12e^x \int_0^x ((e^t - 1)(\ln(1+t))^2 / (2t^3 + 3)) dt}{\alpha^4}$$

$$= \lim_{\alpha \rightarrow 0} 12e^x \left[\frac{(e^\alpha - 1)(\ln(1+\alpha))^2}{4\alpha^3(2\alpha^3 + 3)} \right]$$

$$= 12e^x \left[\frac{(e^\alpha - 1)}{\alpha} \left(\frac{\ln(1+\alpha)}{\alpha} \right)^2 \cdot \frac{1}{4(2\alpha^3 + 3)} \right]$$

$f(x) = e^x$, Now check the option.

11. Ans (B,C,D)

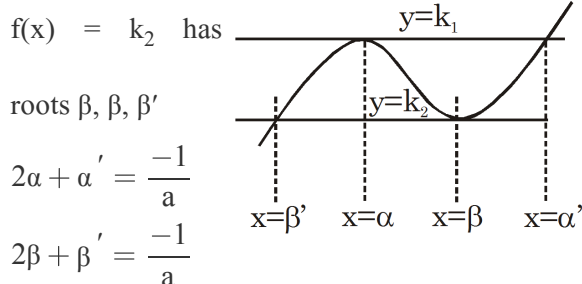
$$f(0) = f(1)$$

$$2 = a + 1 - 2a - 1 + 2$$

$$a = 0 \Rightarrow f(x) = x^2 - x + 2$$

12. Ans (A)

$f(x) = k_1$ has roots α, α, α'



$$2\alpha + \alpha' = \frac{-1}{a}$$

$$2\beta + \beta' = \frac{-1}{a}$$

$$2(\alpha + \beta) + \alpha' + \beta' = \frac{-2}{a} \quad \dots(i)$$

$$f'(x) = 0 \Rightarrow 3x^2 a + 2x - (2a + 1) = 0$$

$$\therefore \alpha + \beta = \frac{-2}{3a} \quad \therefore \alpha' + \beta' = \frac{-2}{a} + \frac{4}{3a} = \frac{-2}{3a}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.50)

$$I = \int \frac{x^{16} \cdot x}{(1+x^2)^{10}} dx \quad 1+x^2 = t$$

$$= \frac{1}{2} \int \frac{(t-1)^8}{t^{10}} dt \quad xdx = \frac{1}{2} dt$$

$$= \frac{1}{2} \int \left(1 - \frac{1}{t} \right)^8 \cdot \frac{1}{t^2} dt \quad \text{Put } 1 - \frac{1}{t} = y$$

$$= \frac{1}{2} \int y^8 dy \quad \frac{1}{t^2} dt = dy$$

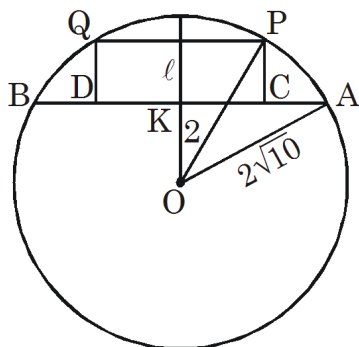
$$= \frac{1}{18} \cdot \left(\frac{x^2}{1+x^2} \right)^9 + C$$

$$\therefore m = 9, n = 18$$

$$\therefore \frac{m}{n} = \frac{1}{2}$$

2. Ans (6.00)

$$AB = 12 \Rightarrow KA = 6$$



Let $PC = \ell$

$$OK = 2$$

$$PQ = 2\sqrt{40 - (2 + \ell)^2}$$

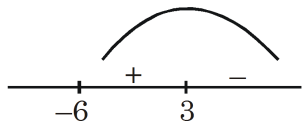
Area of rectangle

$$A = \ell \cdot 2\sqrt{36 - \ell^2 - 4\ell}$$

$$\frac{dA}{d\ell} = 2 \left\{ \frac{\ell(-\ell - 2)}{\sqrt{36 - \ell^2 - 4\ell}} + \sqrt{36 - \ell^2 - 4\ell} \right\}$$

$$\Rightarrow (-\ell^2 - 2\ell + 36 - \ell^2 - 4\ell) \frac{2}{\sqrt{36 - \ell^2 - 4\ell}}$$

$$\Rightarrow -(\ell + 6)(\ell - 3) \cdot \frac{2}{\sqrt{36 - \ell^2 - 4\ell}}$$



$$A_{(\max)} \text{ at } \ell = 3 \Rightarrow A_{\max} = 6\sqrt{15}$$

$$\frac{\lambda}{\sqrt{15}} = 6$$

$$A_{\max} = 6\sqrt{15}$$

3. Ans (8.00)

$$f(x) = \frac{195x(x^2 + 4)}{9x^4 + 97x^2 + 144} = \frac{13(5x)(3x^2 + 12)}{(3x^2 + 12)^2 + (5x)^2}$$

$$= \frac{13}{\frac{3x^2 + 12}{5x} + \frac{5x}{3x^2 + 12}} \text{ for } x \neq 0$$

$$\text{Then } \left| \frac{3x^2 + 12}{5x} \right| \geq \frac{12}{5}$$

$$\text{Hence } |f(x)| \leq \frac{60}{13}$$

$$a - 4b = 8$$

4. Ans (4.58 to 4.59)

$$\int \left(\frac{\sin x \cos x}{1 + \sin x + \cos x} \right)^2 dx = \frac{1}{4} \int \left(\frac{2 \sin x \cos x}{1 + \sin x + \cos x} \right)^2 dx$$

$$\frac{1}{4} \int \left(\frac{1 + 2 \sin x \cos x - 1}{1 + \sin x + \cos x} \right)^2 dx = \frac{1}{4} \int \left(\frac{(\sin x + \cos x)^2 - 1}{1 + \sin x + \cos x} \right)^2 dx$$

$$\frac{1}{4} \int \frac{(\sin x + \cos x + 1)^2 (\sin x + \cos x - 1)^2}{(1 + \sin x + \cos x)^2} dx$$

$$\frac{1}{4} \int (\sin x + \cos x - 1)^2 dx$$

$$= \frac{1}{4} \int (2 + 2 \sin x \cos x - 2 \sin x - 2 \cos x) dx$$

$$\Rightarrow \frac{1}{4} \int (2 + \sin 2x - 2 \sin x - 2 \cos x) dx$$

$$\Rightarrow \frac{x}{2} - \frac{\cos 2x}{8} + \frac{1}{2}(\cos x - \sin x) + \delta$$

$$\Rightarrow \frac{x}{2} - \frac{\cos 2x}{8} + \frac{\cos(x + \frac{\pi}{4})}{\sqrt{2}} + \delta$$

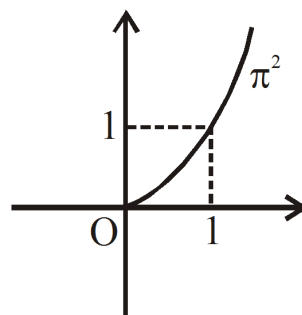
$$\alpha = 2, \beta = -8, \gamma = \sqrt{2}$$

$$|\alpha + \beta + \gamma| = 6 - \sqrt{2} = 6 - 1.414$$

$$= 4.586$$

5. Ans (0.03)

$$f(x) = \max \{x, x^2\} = \begin{cases} x & ; x \in [0, 1] \\ x^2 & ; x \in (1, \infty) \end{cases}$$



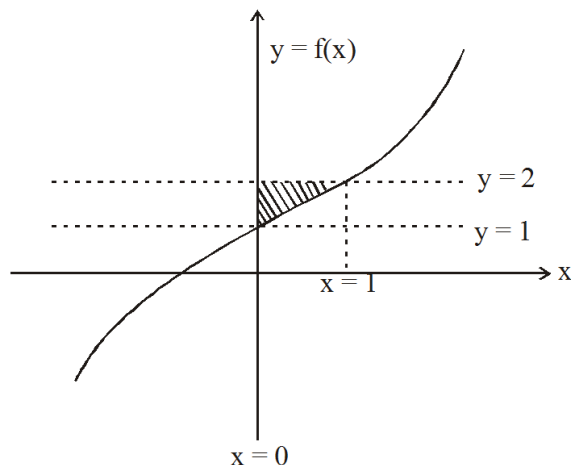
$$\Rightarrow f \circ f(x) = \begin{cases} f(x) & ; 0 \leq f(x) \leq 1 \\ f^2(x) & ; f(x) > 1 \end{cases}$$

$$= \begin{cases} x & 0 \leq x \leq 1 \\ x^4 & x > 1 \end{cases}$$

$$(f \circ f)^{-1}(x) = x^{1/4} \text{ for } x > 1$$

$$\frac{d}{dx} (f \circ f)^{-1}(x) = \frac{1}{4} x^{-3/4} \text{ put } x = 16$$

6. Ans (0.25)



$$A = \int_0^1 2 - (x^3 - 3x^2 + 3x + 1) dx$$

$$A = \int_0^1 (-x^3 + 3x^2 - 3x + 1) dx$$

$$A = -\frac{x^4}{4} + x^3 - \frac{3x^2}{2} + x \Big|_0^1 = \frac{1}{4}$$

7. Ans (3.14)

$$I = \int_1^3 \left[\tan^{-1} \left(\frac{(x-2)^2 - (x-2)}{(x-2)^3 + 1} \right) + \tan^{-1} \left(\frac{1}{(x-2)^2} \right) \right] dx$$

$$= \int_{-1}^1 \left(\tan^{-1} \left(\frac{t^2 - t}{t^3 + 1} \right) + \tan^{-1} \frac{1}{t^2} \right) dt$$

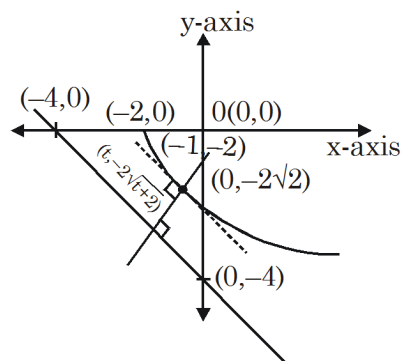
$$= \int_{-1}^1 \left(\tan^{-1} t^2 - \tan^{-1} t + \tan^{-1} \frac{1}{t^2} \right) dt$$

$$= \int_{-1}^1 (\tan^{-1} t^2 - \tan^{-1} t + \cot^{-1} t^2) dt = \pi$$

8. Ans (0.50)

$$\text{Let } C_1 : y = -2\sqrt{x+2} \quad \dots(1)$$

$$C_2 : y = -x - 4 \quad \dots(2)$$



$\therefore$ Diff. b.t.s w.r.t x in (1),

$$\left. \frac{dy}{dx} \right|_{(t, -2\sqrt{t+2})} = -\frac{1}{\sqrt{t+2}} = -1$$

$$\Rightarrow t = -1$$

$$\therefore d = \left(\frac{|-1 - 2 + 4|}{\sqrt{2}} \right)^2 = \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2}$$